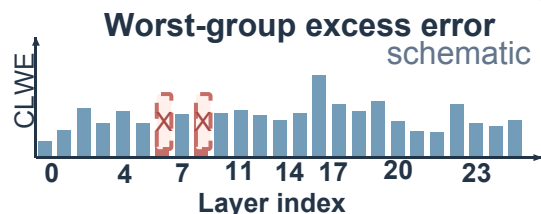
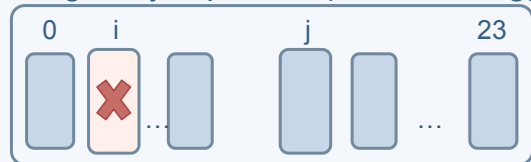


A. CLWE layer selection

EN ZH JA KO

Fine-tuned CosyVoice3
Dense LLM (24 blocks)

Single-layer probes (no retraining)



Rank layers by CLWE

5 0 8 2 ... 20 23

Lower risk Higher risk

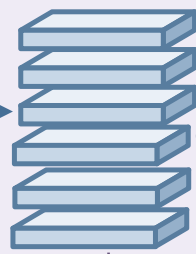
$$\text{CLWE}(i) = \max_g [E_i(g) - E_0(g)]$$

E_0 : dense model E_i : block i removed

B. Structural pruning

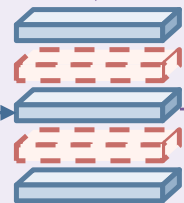
Only the autoregressive LLM is pruned

Dense teacher
Fine-tuned CosyVoice3



Teacher-guided training

Remove k
low-risk blocks



Retain order

Pruned student

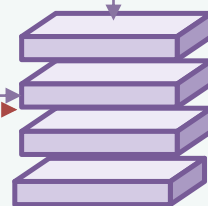
Selected layers

24 $\rightarrow$ 24 - k blocks

Original order preserved

C. Verifier-guided recovery

Text + speech prompt



tokens

Flow + HiFTNet
(fixed)

waveform

Retry

Target text
"... at nine."

Whisper (offline)

WER (EN) / CER (ZH, JA, KO) = 0

"... at five." Reject

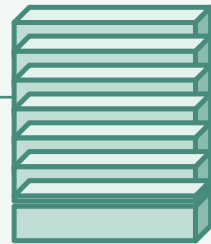
"... at nine." Accept

Accepted speech-token trajectories for recovery



Whisper is excluded from inference

Recovered student tokens

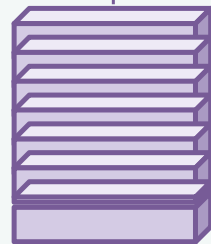


Causal cross-entropy

Update weights

SFT

Fixed topology



Pruned student